BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: V** **Roll No.: TYIT 11** **Date : 01 /08/2026**

Course Name: Big Data Analytics **Page no:**

Practical Number:- 8

Aim :- Implement SVM classification techniques.

Code:

```
From sklearn import datasets
From sklearn.model_selection import train_test_split
From sklearn.svm import SVC
From sklearn.metrics import classification_report, accuracy_score
Import matplotlib.pyplot as plt
Import seaborn as sns

# Load Iris dataset
Iris = datasets.load_iris()
X = iris.data
Y = iris.target
# Split into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.3, random_state=42
)

# Initialize and train the SVM model
Model = SVC(kernel='linear') # You can use 'rbf', 'poly', or 'sigmoid'
Model.fit(X_train, y_train)

# Make predictions
Y_pred = model.predict(X_test)

# Evaluate the model
Print("Ravit TYIT-11")
Print("Accuracy:", accuracy_score(y_test, y_pred))
Print("\nClassification Report:\n", classification_report(y_test, y_pred))

# (Optional) Confusion matrix visualization
From sklearn.metrics import confusion_matrix
Import numpy as np

Cm = confusion_matrix(y_test, y_pred)
Sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
        Xticklabels=iris.target_names,
        Yticklabels=iris.target_names)

Plt.title("TYIT-11 Confusion Matrix")
Plt.xlabel("Predicted")
Plt.ylabel("Actual")
Plt.show()
```

Teacher's Signature

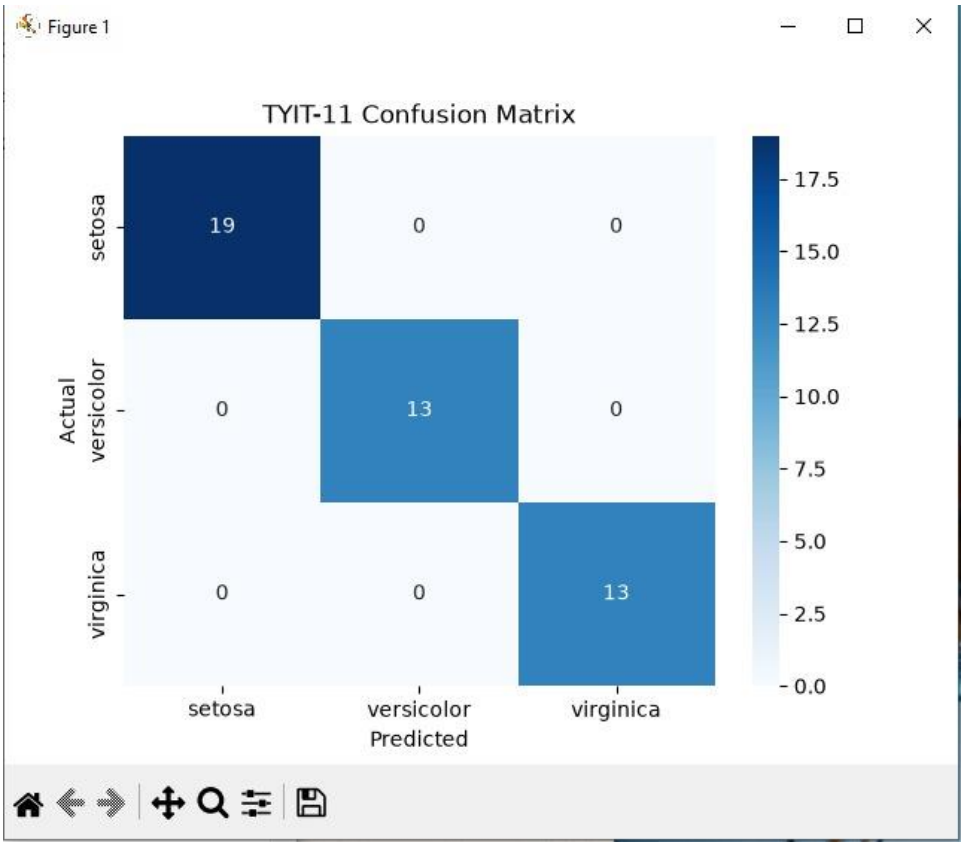
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Output :




```
Ravit TYIT-11
Accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        19
           1       1.00      1.00      1.00        13
           2       1.00      1.00      1.00        13

    accuracy                           1.00        45
   macro avg       1.00      1.00      1.00        45
weighted avg       1.00      1.00      1.00        45

>>>
```

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